

Dr. Dipankar Bhattacharya, Ph.D.

Marie Skłodowska-Curie Postdoctoral Fellow (2025–2027)
Dyson School, Imperial College London, United Kingdom

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Research Profile

Independent robotics researcher developing model-based and learning-enabled robotic systems for healthcare, rehabilitation, and advanced manufacturing. My work integrates soft robotics, cable-driven actuation, deformable-object manipulation, perception, and control, including industry-funded esophageal stent testing on the RoSE platform (BvW Holding, NZD 39,000, 2018). As Marie Skłodowska-Curie Fellow at Imperial College London, I lead CASREx (Grant No. 101205678; EUR 276,187.92, 24 months) on cable-driven exosuits for upper-arm muscle-impairment rehabilitation, building on RoSE (Soft Robotics, 2021), cable wrapping modeling (MMT, 2025; IEEE TCST, 2023), and fabric manipulation (IEEE IROS, 2026; IEEE RA-L, 2025).

Education

- 2016 – 2021 ■ **Ph.D., Mechatronics Engineering** in Department of Mechanical and Mechatronics Engineering, The University of Auckland (UoA), 01/08/2016 – 14/04/2021.
Thesis title: *Robotic Soft Esophagus Design, Modelling, and Control for Endoprosthesis Stent Testing and Food Swallow Investigation*
Developed RoSE for physiologically relevant stent testing and swallow investigation; outputs: 3 journal articles (2 in Soft Robotics, 2 IEEE TIE), 2 conference papers, 1 invited book chapter (Springer, 2019).
- 2011 – 2013 ■ **M.Tech, Systems and Control** in Department of Electrical Engineering, Indian Institute of Technology (IIT), 22/07/2011 – 04/07/2013.
Dissertation title: *Non-Linear Predictive Control Model using Relevance Vector Machine*. GPA: **8.4 / 10.0**
Nonlinear MPC with Relevance Vector Machine, validated on a magnetic-levitation system; 1 journal article (Int. J. Sustainable Energy, 2015).
- 2004 – 2010 ■ **B.Tech, Electronics and Communication Engineering**, North Eastern Regional Institute of Science and Technology (NERIST), 21/07/2004 – 23/06/2010.
Project title: *Design and Synthesis of a Sixth Order Tow Thomas Bi-quad filter*. GPA: **9.1 / 10.0**

Employment History

- 2025 – Present ■ **Marie Skłodowska-Curie Postdoctoral Fellow**, Dyson School of Design Engineering, Imperial College London. Supervisor: Prof. Thrishantha Nanayakkara (Morph Lab).
CASREx: Cable-driven Arm Soft Reconfigurable Exoskeleton for Muscle-Impairment Rehabilitation (Grant No. 101205678; EUR 276,187.92, 24 months).
 - Leading fellowship project on human-inspired stiffness modulation for cable-driven elbow exosuits for upper arm muscle impairment rehabilitation.
 - Integrating human trials, EMG, digital twins, and reinforcement learning for adaptive exosuit control.
 - Developing and validating a cable-driven manipulator-exosuit platform with compliant actuation and antagonistic co-contraction control.

Employment History (continued)

- 2026 – Present ■ **Assistant Supervisor**, Dyson School of Design Engineering, Imperial College London, United Kingdom, 15/01/2026 – Present.
Assistant Supervisor for two Year 3 MEng (Design Engineering) placement students on exosuit R&D (March 2026–Present).
Delivered lecture on **Robotics Simulation and Control using PyDrake** for *DESE71002 Robotics Research Projects*, Dyson School of Engineering, Imperial College London (28/01/2026).
- 2024–2025 ■ **Senior Research Engineer–Control Systems** in the Center for Transformative Garment Production (TransGP), Mentor: Prof. Kazuhiro Kosuge, 02/07/2024 – 31/10/2025.
Physics-based Cloth Manipulation with policy generated from imitation learning.
 - Led multidisciplinary work on dual-arm fabric pre-sewing handling and flattening (mesh perception + imitation learning).
 - Contributed to IEEE IROS (2026) on dual-arm fabric flattening (RTFF), IEEE RA-L (2025) on automated straight-line sewing, and IEEE ICMA (2025) fabric destacking.
 - Built and validated robotic fabric manipulation testbeds for industrial sewing and destacking workflows (May–Oct 2025).
- 2024 – 2025 ■ **Visiting Research Associate** in the Department of Electrical and Electronic Engineering, Hong Kong University (HKU), 02/07/2024 – 31/10/2025.
- 2021 – 2024 ■ **Postdoctoral Fellow** in the Department of Mechanical and Automation, The Chinese University of Hong Kong. Supervisor: Prof. Darwin Lau, 01/03/2021 – 30/06/2024
Project title 1: Modeling and control of cable-driven robots with cable wrapping around obstacles and links.
Project title 2: Design of a cable-driven growing vine robot for safer false ceiling inspections.
 - Authored journal papers on cable wrapping (MMT, 2025) and iterative-learning control of hybrid cable-driven robots (IEEE TCST, 2023).
 - Mentored undergraduate final-year and placement students, including a cable-driven vine-robot inspection project (CUHK, 2023–2024).
 - Co-authored grant proposals and technical reports; mentored PhD students at HKU and CUHK (2021–2026).
- 2021 – 2022 ■ **Visiting Postdoctoral Fellow** in Laboratoire des Sciences du Numérique de Nantes (LS2N), École Centrale de Nantes (ECN). Supervisors: Prof. Stephane Caro and Prof. Darwin Lau, 26/10/2021 – 05/01/2022.
Project title: Iterative-learning control of cable-driven parallel robot.
Delivered iterative-learning control experiments on cable-driven parallel robots (Oct 2021 – Jan 2022).
- 2017 – 2020 ■ **Graduate Teaching Assistant**, The University of Auckland, New Zealand, 2017–2020.
- 2018 – 2019 ■ **Research Assistant**, The University of Auckland (Prof. Peter Xu), 09/01/2018 – 08/01/2019.
 - Measured migration of esophageal stents of multiple designs on robotics soft esophagus platform (BvW-funded project), and published in Soft Robotics (2021).
 - Developed actuation protocols and test reports for industry stent-testing deliverables.
- 2013 – 2016 ■ **Lecturer (Assistant Professor Grade-1)**, School of Electrical, Electronics and Communication Engineering, Galgotias University, India, 19/07/2013–18/06/2016.

Teaching and Supervision

Teaching

- 2026 ■ **DESE71002**, *Robotics Research Projects*, Year 4/MSc (Design Engineering), Dyson School of Design Engineering, Imperial College London, United Kingdom.
Guest lecture on *Robotics Simulation and Control using PyDrake* (28/01/2026).
- 2021 – 2024 ■ **MAEG5090**, *Topics in Robotics*, for postgraduate students (MSc/MPhil), Department of Mechanical and Automation Engineering, The Chinese University of Hong Kong, Hong Kong SAR. Class size: 50 students.
- 2017 – 2020 ■ **MECHENG 706**, *Mechatronics Design Projects*, for 4th year undergraduate students, The University of Auckland, New Zealand.
- **MECHENG 306**, *Design of Sensing and Actuating Systems*, for 3rd year undergraduate students, The University of Auckland, New Zealand.
- **ENGGEN 115**, *Principles of Engineering Design*, for undergraduate students, The University of Auckland, New Zealand.
- **ENGGEN 131**, *Introduction to Engineering Computation and Software Development*, for undergraduate students, The University of Auckland, New Zealand.
- 2013 – 2016 ■ Undergraduate courses (class size: 60 each), School of Electrical, Electronics and Communication Engineering, Galgotias University, India:
BECE3304 *Neural Networks and Fuzzy Control*; **BECE2010** *Digital Electronics*; **BECE2016** *Signals and Systems*; **BTEE2002** *Network Analysis and Synthesis*; **BEEE3002** *Control Systems*; **BEEE2001** *Electrical Measurement and Instrumentation*; **BECE3020** *Digital Signal Processing*; **BTEE3019** *Advanced Control System*.
- Postgraduate courses (class size: 20), School of Electrical, Electronics and Communication Engineering, Galgotias University, India:
BEEE5004 *Digital Control*.

Supervision

- 2026 – Present ■ **Assistant Supervisor**, Dyson School, Imperial College London.
Year 3 MEng Design Engineering placement: **Ben Surtees** and **Lijiangke Liu** (from March 2026). Exosuit R&D.
- 2024 – 2026 ■ **PhD research supervision**, University of Hong Kong.
 - **Wenbo Dong** (2024–2026) — fabric robotics, perception, and machine learning.
 - **Kai Tang** (2024–2026) — dual-arm fabric manipulation; co-authored RTFF / IROS 2026.
 - **Bingchen Jin** (2024–2025) — automated garment sewing; co-authored IEEE RA-L 2025.
- 2023 – 2024 ■ **Herman** — Year 4 BEng placement, Mechanical and Automation Engineering, The Chinese University of Hong Kong.

Research Publications

Journal (Published/Accepted)

- 1 Wang, W., Bhattacharya, D., Kobayashi, A., Seino, A., Tokuda, F., Tien, N., & Kosuge, K. (2026). Automated Fabric Alignment Using Global-Local Optimization and Point Cloud Registration [accepted]. *IEEE Transactions on Automation Science and Engineering*. <https://doi.org/10.36227/techrxiv.175691617.74372270>
- 2 Jin*, B., Kobayashi*, A., Bhattacharya, D., Seino, A., Tokuda, F., Tien, N., & Kosuge, K. (2025). Automated Straight-line Sewing of Stretchable Fabrics with Different Lengths [*Equal contribution.]. *IEEE Robotics and Automation Letters*, 1–8. <https://doi.org/10.1109/LRA.2025.3619791>
- 3 Bhattacharya, D., Cheung, T. K., Wang, Y., & Lau, D. (2025). Kinematic and Dynamic Modeling of Cable-Object Interference and Wrapping in Complex Geometrical-Shaped Cable-Driven Parallel Robots. *Mechanism and Machine Theory*, 214, 106092. <https://doi.org/10.1016/j.mechmachtheory.2025.106092>
- 4 Bhattacharya, D., Chan, Y. P., Shang, S., Chan, Y. S., Tan, Y., & Lau, D. (2023). Tri-Space Operational Control of Redundant Multilink and Hybrid Cable-Driven Parallel Robots Using an Iterative-Learning based Reactive Approach. *IEEE Transactions on Control Systems and Technology*. <https://doi.org/10.1109/TCST.2023.3263689>
- 5 Bhattacharya, D., Cheng, L. K., & Xu, W. (2021). Nonlinear Model Predictive Control of a Robotic Soft Esophagus for Endoprosthetic Stent Testing. *IEEE Transactions on Industrial Electronics*. <https://doi.org/10.1109/TIE.2021.3121755>
- 6 Bhattacharya, D., Jesna, S., Cheng, L. K., & Xu, W. (2021). RoSE: A Robotic Soft Esophagus for Endoprosthetic Stent Testing. *Soft Robotics*. <https://doi.org/10.1089/soro.2019.0205>
- 7 Dang, Y., Liu, Y., Hashem, R., Bhattacharya, D., Allen, J., Stommel, M., Cheng, L. K., & Xu, W. (2021). SoGut: A Soft Robotic Gastric Simulator. *Soft Robotics*. <https://doi.org/10.1089/soro.2019.0136>
- 8 Bhattacharya, D., Cheng, L. K., & Xu, W. (2020). Sparse Machine Learning Discovery of Dynamic Differential Equation of an Esophageal Swallowing Robot. *IEEE Transactions on Industrial Electronics*, 67(6), 4711–4720. <https://doi.org/10.1109/TIE.2019.2928239>
- 9 Bhattacharya, D., Nisha, M. G., & Pillai, G. (2015). Relevance vector-machine-based solar cell model. *International Journal of Sustainable Energy*, 34(10), 685–692. <https://doi.org/10.1080/14786451.2014.885030>
- 10 Bhattacharya*, D., Tang*, K., Hang, X., Tokuda, F., Tien, N., & Kosuge, K. (2026). RTFF: Random-to-target fabric flattening policy using dual-arm manipulator [accepted. *Equal contribution]. *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.
- 11 Dong, W., Bhattacharya, D., Kobayashi, A., Seino, A., Tokuda, F., Huang, X., Tang, K., Tien, N. C., & Kosuge, K. (2025). Precise Top-layer Fabric Segmentation for Fabric Destacking with Edge- and Shape-Aware Deep Networks. *2025 IEEE International Conference on Mechatronics and Automation (ICMA)*, 1343–1348.
- 12 Bhattacharya, D., Cheng, L. K., & Xu, W. (2018). Testing and Analysis of Migration Displacement of a Flared Stent Deployed in an Esophageal Swallowing Robot. *25th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*, 1–6. <https://doi.org/10.1109/M2VIP.2018.8600911>
- 13 Bhattacharya, D., Cheng, L. K., Dirven, S., & Xu, W. (2017). Actuation planning and modeling of a soft swallowing robot. *24th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*, 1–6. <https://doi.org/10.1109/M2VIP.2017.8211476>

Books and Chapters

- 1 Bhattacharya, D., Cheng, L. K., Dirven, S., & Xu, W. (2019). *Artificial intelligence approach to the trajectory generation and dynamics of a soft robotic swallowing simulator*. Springer International Publishing. https://doi.org/10.1007/978-3-319-78452-6_1

In Review/Preparation

- 1 Wang, Y., Chow, E., Bhattacharya, D., & Lau, D. (2026). Dual-Stage Spatial and Continuum Cable-Driven Parallel Robot for Airlifting in Bored Pile Construction [in preparation].

Grants and Scholarships

Grants/Fellowships

- 2025  Recipient, **Horizon Europe Marie Skłodowska-Curie Postdoctoral Fellowship** for the project “CASREx” (Grant No. 101205678), hosted at **Imperial College London**. Funded under HORIZON-MSCA-2024-PF-01-01. Duration: 24 months; value: **EUR 276,187.92**.
- 2023  Co-author, Innovation and Technology Fund (ITF) Partnership Research Programme (PRP) grant for the project “Flexible Robotic Palm for Human Comforting Touching Sense.”, hosted at the Chinese University of Hong Kong (CUHK).
- 2022  Co-authored Research Talent Hub for Innovation and Technology Fund (RTH-ITF) grant for the project “Hybrid Cable-Driven with Active Soft Backbone Continuum Robot for Dexterous Manipulation in Service Applications.”
- 2018  Recipient, Dynamic Testing of Implant Stents Using Robotic Soft Esophagus (RoSE), BvW Holding, Cham. Duration: 12 months; value: **NZD 39,000**.
- 2018, 2019  Recipient, Riddet Travel Grant, Riddet Institute Centre of Research Excellence (NZD 3,000).

Scholarships

- 2019–2020  Recipient, Doctor of Philosophy Scholarship, University of Auckland.
- 2016–2019  Recipient, Doctor of Philosophy Scholarship, Riddet Institute Centre of Research Excellence.
- 2011–2013  Recipient, Master of Technology Scholarship, Ministry of Human Resource Development (MHRD), Government of India.

Academic Awards, and Miscellaneous Achievements

Academic Awards

- 15/12/25  Selected, **European Talent Academy 2026** (Imperial College London, TU Munich, Politecnico di Milano).
- 2017  Recipient, **Best Conference paper award**, 5th International Conference on Robotic Intelligence Technology and Applications (RiTA), Daejeon, Korea.
-  **Ranked 2nd** in poster presentation in Delivery of Functionality in Complex Food Systems (DoF 2017) Conference.

Competitive Examinations

- 2011  **Percentile: 99.12% (Rank 1,211/1,37,853)** in Graduate Aptitude Test of Engineering (GATE).
- 2010  **Percentile: 96.00%** in Graduate Aptitude Test of Engineering (GATE).
- 2004  **Ranked 31st** in All India Junior mathematics Olympiad.

Professional Activities

- 05/02/26  Invited Speaker, **Dyson Engineering Departmental Seminar**, Imperial College London.
- 29/01/26  Invited Speaker, **Imperial Robotics Postgrad Committee**, Imperial College London. Talk: *Learning-based Fabric Manipulation with Mesh Perception and Visual Servoing*.
- 28/01/26  Delivered lecture on **Robotics Simulation and Control using PyDrake** for DESE71002 Robotics Research Projects, Dyson School of Engineering, Imperial College London.
- 08/08/25  Invited Speaker, **World Robotics Conference (WRC 2025)**, Beijing, China. Talk: *Fabric modeling and perception for garment manufacturing*.
- 03/08/25  Workshop Organizer and Presenter, **IEEE ICMA 2025**, Beijing, China. Talk: *Mesh-based Fabric State Estimation and Manipulation*.

Academic Awards, and Miscellaneous Achievements (continued)

29/05/23	■	Poster Presentation, IEEE ICRA 2023 , ExCeL London, United Kingdom.
2018	■	Presenter, BioMouth 2018 Meet , Auckland, New Zealand.
2021 - Current	■	Reviewer of IEEE Robotics and Automation Letters (RAL), IEEE International Conference on Robotics and Automation (ICRA), ASME Journal of Mechanisms and Robotics (JMR).
2017 - Current	■	Member of the Institute of Electrical and Electronics Engineers (IEEE).
2014	■	Former member of the Electrical Engineering Society (EES), IIT Roorkee, and NERIST Electronics Society (NES), NERIST.
2006	■	Have an 'Outstanding Certificate' in National Service Scheme (NSS) for contributing in activities like blood donation camps, village cleanliness programmes and children education.

Skills

Languages	■	English and Hindi: professional reading, writing, and speaking proficiency.
Research areas	■	Soft robotics, cable-driven robots, robotic perception, deformable-object manipulation, digital twin and control, learning-based modeling and control, and robotic systems for healthcare and manufacturing.
Programming	■	Python, MATLAB, PyTorch, ROS2, LabVIEW, assembly-level programming (8085, 8086, MASM32, and 8051), Git, and LaTeX.
Machine learning	■	Imitation learning, reinforcement learning, sparse identification modelling, nonlinear MPC with learned dynamics, mesh-based deep perception for deformable objects.
Simulation	■	NVIDIA Isaac Sim, pydrake, LeRobot, OpenSim
CAD and others	■	Onshape, Blender, SOLIDWORKS, Autodesk Inventor, Inkscape, and Lightroom.
Hardware	■	NVIDIA Jetson, Raspberry Pi, myRIO, and Arduino.

References

Prof Kazuhiro Kosuge

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